MIL-H-85330A(AS)
17 April 1981
Superseding
MIL-H-85330(AS)
18 July 1979

MILITARY SPECIFICATION

HYBRID TEST SYSTEM

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the minimum performance and design characteristics for a Hybrid Test System (HTS). The HTS will be used for functional and diagnostic test of analog and hybrid Units Under Test (UUTs) within a variety of weapon systems either currently in inventory or projected for subsequent introduction into inventory.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-7793	Time Totalizing Meter
MIL-Q-9858	Quality Program Requirements
MIL-T-18303	Test Procedures; Preproduction and Inspection for Aircraft Electronic Equipment, format for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment; General Specification for
MIL-S-52779	Software Quality Assurance Program Requirements
MIL-D-85424	Performance and Interface Specification for Digital Word Generator (Enhanced)

Standards

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-210	Climactic Extremes for Military Equipment
MIL-STD-461	Electromagnetic Interference Characteristic Requirements for Equipment
MIL-STD-470	Maintainability Program Requirement (for Systems and Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-794	Parts and Equipment, Procedures for Packing and Packaging of
MIL-STD-847	Reliability Program for Systems and Equipment Development and Production
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-1399	Interface Standards for Shipboard Systems
MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-2076	Unit Under Test Compatibility with ATE, General Requirements for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

Military

OPNAVINST 4790.2B CH-7, VOLII,(A)717	Naval Aviation Logistics Command Management Information System (NALCOMIS)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI X3.9-1978	American National Standards American National Fortran
ANSI X3.4	American Standard Code for Information Interchange (ASCII)

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, N.Y. 10019.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE-STD-416-1978	IEEE/ARINC Standard ATLAS Test Language and ATLAS Syntax
IEEE-STD-488-1978	IEEE Standard Digital Interface for Programmable Instrumentation

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, N.Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 HTS definition. The HTS shall be a computer controlled automatic test set which will provide the capability for performing functional and diagnostic fault isolation tests on analog or hybrid UUTs. The HTS, when used with appropriate Test Program Sets (TPSs) (not covered as part of this specification), shall provide the capabilities to rapidly and accurately perform the following functions.

a. Test a UUT in a manner simulating its normal system environment to detect faults which would preclude satisfactory system performance.

b. Isolate the cause of UUT failure.

c. Perform self-test on the HTS to determine if it is functionally capable of testing UUTs.

d. Isolate HTS malfunction(s) to the next lower HTS replaceable unit.

e. Perform self-certification on the HTS in accordance with the requirements of 3.2.7.5 contained herein.

f. Direct recalibration of the HTS resources in accordance with the requirements of 3.2.7.5 contained herein.

3.2 General design requirements.

3.2.1 Selection of hardware. The hardware selected for incorporation into the HTS shall be state-of-the-art, currently manufactured and of a proven design such that performance, reliability and accuracy are readily verifiable. Selected hardware shall be common with, or have optimum logistics commonality to successfully demonstrated implementations of ATE elements currently existing within ATE in the inventory of the procuring activity. Hardware meeting these requirements which has been built to best commercial practice is acceptable.

3.2.1.1 Hardware built to best commercial practice. Hardware built to best commercial practice shall be that hardware which is:

a. Priced and offered for public sale

b. Developed and tested to published criteria

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

c. Described as to form, construction and performance by available technical data.

d. Constructed such that spare parts are commercially available

e. Independently perfected by the manufacturer

f. Upwards compatible in an applicable product line of the manufacturer

3.2.2 Design and construction. Detailed mechanical and electrical design of the system shall be accomplished such that the system is suitable for use in Intermediate and Depot Level military maintenance activities, either ashore or afloat. Mechanical and electrical design of the system may be accomplished in accordance with best commercial practices consistent with the HTS capabilities specified herein. Selections of HTS modular hardware elements shall take into consideration the test compatibility of these elements with provisions delineated within MIL-STD-2076.

3.2.2.1 Total weight. The maximum total weight of the HTS system, including cables, shall be 4,000 lbs.

3.2.2.2 Size. The HTS station shall conform to the following:

a. Height: 78-1/2 inches maximum (exclusive of mounting hardware and provisions for hoisting)

b. Depth: 38 inches maximum (exclusive of work surface and patch panel)

c. Width(overall): Shall be 96-1/4 inches maximum

3.2.2.3 Work surface. A nonconductive work surface area of not less than four (4) square feet extending from the front of the station shall be provided. The work surface shall be approximately 36 inches above the floor and shall be capable of supporting 200 pounds at its outer edge.

3.2.2.4 Modularity. The HTS shall be a fully integrated and operational system comprised of modular ATE hardware elements for ease in implementing station modification or reconfiguration requirements. The modular design of the system shall be such that interchanging of modular system elements with functionally equivalent units will not modify or degrade system performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.2.2.5 Time totalizing meter. A Time Totalizing Meter displaying total system operating time shall be provided. Its location shall be such that it may be easily read without need for cover removal from the equipment. The time totalizing meter shall be compatible with the requirements of MIL-M-7793.

3.2.2.6 Cables and Connectors. The system shall employ subassemblies, cables, interconnected harnesses and connectors of uniform design and physical dimensions. Connector locations shall be such that the connecting cables will not interfere with the operating controls. Constraints in meeting their requirements by the use of commercial equipment, shall be reported to the contracting officer with alternative recommendations for his deposition and pertinent direction to the contractor.

3.2.2.7 Material mechanical properties. The allowable mechanical properties of material used in the design of the system shall be in accordance with best commercial practice. Mechanical properties of materials used in design shall be obtained in accordance with best commercial pratices. In all cases, the most conservative (minimum) values of material strength parameters shall be assumed for use in the design process.

3.2.2.8 Design stresses. The system shall be of adequate strength to withstand, without damage, stresses incident to moving, handling in transit, hoisting, and tie down aboard transporting vehicles.

3.2.2.9 Growth. The system shall include space and provisions for a 10% increase in contained equipment without exceeding the size limitations cited herein. Connectors shall contain sufficient spare pins to achieve this growth.

3.2.3 Computer interface. The interface between peripherals, stimulus, measurement, and switching units and the station computer shall meet the requirements of IEEE STD-488-1978 (subject to commercial availability and conflicting requirements for high speed data transmission).

3.2.4 Reliability. The contractor shall conduct a reliability program compatible with the requirements of MIL-STD-847 and generate the reports and data listed in 6.2.2.

3.2.4.1 Useful life. The useful life of the system shall not be less than ten (10) years under any combination of operating life and storage life without exceeding the operational service life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.2.4.2 Operating Life. The system shall be designed to have a minimum total operating life of 50,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and wearout life of such parts shall be determined by the contractor and their replacement shall be documented in maintenance publications used in conjunction with the system.

3.2.4.3 Storage life. The system shall have a storage life of up to 30 months under the environmental operating conditions contained herein or under conditions specified in MIL-STD-210, when packaged in accordance with MIL-STD-794 Level A. Upon being withdrawn from storage and placed in service, the system shall meet the requirements invoked herein for no less than the total specified operating life.

3.2.4.4 Reliability in mean time between failures (MTBF). The system MTBF shall not be less than 500 hours.

3.2.5 Maintenance times. The system shall have a Mean Time to Repair (MTTR) of 0.5 hours or less where MTTR is the mean value of elapsed unscheduled maintenance times to detect, troubleshoot, and repair system malfunctions. The maximum time for any discrete repair at the 95 percentile shall be 1.5 hours.

3.2.6 Electromagnetic interface. The system shall operate satisfactorily in a maintenance shop environment, and shall not degrade nor be degraded by equipment located three (3) feet or greater from the tester and meeting MIL-STD-461.

3.2.7 System maintenance. A maintainability program compatible with MIL-STD-470 shall be utilized. The system shall be designed in accordance with the following maintenance requirements. In addition, a maintainability demonstration in accordance with MIL-STD-471 shall be performed.

3.2.7.1 Access. The system shall be constructed such that all maintenance can be accomplished from the front of the test station.

3.2.7.2 Extended operation of electronic assemblies (Drawers). The system shall be capable of operation with any or all of the electronic drawers (major assemblies) in the extended position, to facilitate maintenance. The use of auxiliary drawer extenders shall not be required to achieve this capability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.2.7.3 Scheduled maintenance requirements. Except for the cleaning or replacement of air filters, and periodic maintenance of the disc drive(s), the system shall not require scheduled maintenance. Calibration of instruments and power supplies included in test station shall not be considered periodic maintenance.

3.2.7.4 Self-test and self-test documentation. The HTS shall be designed in a manner that facilitates rapid and effective self-test. In addition, the contractor shall prepare documentation for built-in-test, system confidence test, system self-test, system self-test adapter test program and the procedure required in c.5. below. Self-test features to be included in the HTS design include the following:

a. Built-in-Test. Built-in-Test (BIT) features of the HTS will, at a minimum, include:

(1) Computer Self-Check
(2) Memory Checks (sum checks)
(3) Data Transfer Checks (parity)
(4) Power Checks
(5) Safety Checks
(6) Heat Monitor Checks

Failure of a BIT test shall be automatically indicated to the operator in a manner that will allow him to recognize the source and nature of the failure.

b. System confidence test. The system confidence test shall be designed to test all basic system functions at selected ranges and scales. The system confidence test shall be designed to execute in ten (10) minutes or less.

c. System self-test. The HTS shall be designed in a manner that facilitates the performance of a comprehensive self-test.

(1) The self-test program shall, as a minimum, be designed such that all ranges and functions of each test station instrument, as well as all system switch paths are exercised under program control.

(2) A self-test adapter to be connected to the system interface shall be designed to facilitate the performance of the system self-test. A self-test program for the adapter shall also be generated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

(3) Isolation provided by the design of the self-test program in conjunction with the HTS design shall automatically isolate the cause of an HTS malfunction to the single commercial building block or the lowest replaceable subassembly for which repair is accomplished by the replacement of piece parts. This low level of ambiguity shall be made possible by design of the system self-test program on a pyramid of confidence basis wherein proven system elements are used to verify the performance of a single subsequent system element.

(4) Design of the system self-test program shall, to the greatest extent possible, be segmented to allow for operator selection of one (1) or a series of system elements to be tested should a full system test not be desired.

(5) A procedure for isolation by automatic, semi-automatic, and manual means shall be generated and utilized in the event that a test station malfunction precludes the execution of the self-test program.

3.2.7.5 Calibration. The system shall operate as herein specified for a period of not less than nine (9) months without requiring calibration of any of its equipment. A system certification program shall be generated to verify that instruments are performing to specified accuracies at the system interface. The system certification program shall employ a minimum number of external equipments.

a. Instrument calibration shall be performed only when the instrument fails the system certification test (periodic calibration not required). Certification shall be accomplished on station to the maximum extent possible.

b. To facilitate automatic system certification test, the system design shall include the incorporation of secondary standards within the system to act as a reference source. At a minimum, a voltage and frequency standard will be required.

3.2.7.6 Maintenance Management Information System (MMIS). The HTS shall include a Maintenance Management Information System (MMIS) which shall provide for automatic prompting, operator request, operator entry, HTS disc storage, display and listing of HTS maintenance data in a standard format.

a. HTS resident maintenance data shall minimally consist of ATE related data elements addressed within the Naval Aviation Logistics Command Management Information System (NALCOMIS) (Reference OPNAVINST 4790.2B, CH-7, VOL II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

b. HTS resident data elements shall be catalogued to allow ease in data manipulation/entry by programs which will evaluate and track HTS equipment performance and reliability, and record and feedback failure trend and/or other analyses to optimize control of assets.

c. This specification includes the MMIS but does not include the data manipulation programs which may be acquired/developed at a later date.

3.2.8 Nameplates. The test station shall be identified with standard nameplates of commercial format. Location of nameplates shall be coordinated with and approved by the procuring activity. Nomenclature assignment shall be accomplished in accordance with MIL-N-18307.

3.2.9 Service conditions. No fixed part or assembly shall become undesirably free or sluggish in operation, no movable part or control shall be shifted in setting, position, or adjustment, and there shall be no performance degradation during operation after storage or use of the system in the ambient conditions specified in the following paragraphs.

3.2.9.1 Altitude.

a. Non-Operating:	The equipment shall withstand altitudes from 0 to 50,000 feet MSL (30.0 to 3.4 inches Hg), unless otherwise specified herein.
b. Operating:	The equipment shall withstand altitudes from 0 to 5,000 feet MSL (30.0 to 25.0 inches Hg).

3.2.9.2 Shock and vibration. The equipment shall not suffer damage or fail to provide specified performance when subjected to shock or continuous vibration along each of three (3) mutually perpendicular axes within the range of 10-33 HZ at 0.010 inches peak-to-peak excursion.

3.2.9.3 Humidity. The equipment shall withstand the effects of ambient air relative humidity up to 80 per cent (non-condensing) within the specified operating and non-operating temperature ranges, unless otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.2.9.4 Temperature.

a.	Non-Operating:	The equipment shall withstand temperatures from -20°C to +65°C, unless otherwise specified herein.
b.	Operating:	The equipment shall withstand temperatures from +10°C to +35°C.

3.2.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under conditions of laboratory checkout, acceptance testing and use of the system during Intermediate and Depot Level maintenance:

a.	Temperature:	20 ±5°C
b.	Altitude:	Normal sea level barometric pressures
c.	Humidity:	75 per cent relative humidity (non-condensing) in the ambient air (shop) environment
d.	Input Power:	See 3.2.12

3.2.11 Transportability. The system shall be transportable by all modes of transportation to permit employment, deployment, and logistic support. The system shall provide ease in handling, loading, and securing. Provisions shall be made on the cabinets for lifting eyes or other suitable hoisting accommodations. The HTS shall be transportable in accordance with MIL-T-28800.

3.2.12 Primary input power requirements. The equipment shall meet specified performance requirements when supplied with 3-phase power, 115 VRMS, 60 HZ, Delta connected, 3 wire (plus safety ground) per MIL-STD-1399. By using minimal external adapter equipment the system shall operate from the following power source: 3-phase, 115 VRMS, 60 HZ, Wye connected, 4 wire (plus safety ground), grounded neutral per MIL-STD-1399. Input power required by the equipment shall be specified by the offeror.

3.2.12.1 Voltage protection. The equipment shall withstand without permanent damage any overvoltage or undervoltage condition, momentary or otherwise. The equipment shall also be protected against transients on the primary input power source. Transient suppression shall be provided such that transients, including equipment initial turn on transients, shall be clipped for both positive and negative peak voltage excursions above and below a 150 VRMS envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.2.13 Overload protection. All parts and circuits of the equipment which are likely to carry an overload due to failure or poor adjustment shall be designed to withstand such overload without permanent damage, or shall have the necessary fuses, circuit breakers or current limiting circuitry incorporated.

3.2.14 Safety. The HTS System shall be designed such as to preclude incorporation of features resulting in physical and electrical hazards to operating/maintenance personnel and equipment. The system shall be designed so that a failure in one area will neither cause damage in another area of the system nor in the UUT. In the event of an electrical or thermal overload, provisions shall be incorporated in the equipment to automatically remove power. A safety program in accordance with MIL-STD-882 and personnel safety considerations in accordance with MIL-T-28800 shall be utilized. A Systems Safety Program Plan shall be developed by the contractor and provided for approval. The contractor shall also be responsible for preparation of safety status reports and hazard analysis reports.

3.2.15 Warm-up time. The HTS system shall become fully operational within twenty (20) minutes of power turn-on.

3.2.16 Grounding. Ground loops and common ground returns for signal and power grounds shall be avoided. The design of the ground system shall protect operating and maintenance personnel from electrical hazards, provide the necessary shielding for signal circuits, and minimize EMI. Primary AC power returns (neutral) shall remain isolated from all signal and chassis grounds.

3.2.17 Power distribution and control. The HTS system shall contain a Power Distribution and Control panel for primary input power. Offeror shall describe the routing and control of HTS input power, the location of the panel in the system, panel components (power-on indicators, etc.) and any associated HTS elements (rack circuit breakers, etc. The HTS shall provide for status indications pursuant to 3.2.7.4

3.2.18 First Article Tests. The contractor shall furnish sample units for first article inspection and approval (see 4.3.a and 6.2.4).

3.3 Detail design requirements. Unless otherwise specified, all specifications provided herein are system requirements as measurable on universal and dedicated pins (Ref. 3.3.5) at the ATE/UUT system interface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.1 Control subsystem. The Control Subsystem shall provide the necessary computer control of station assets. It shall consist of:

a. Digital Computer

b. Disc Drive(s)

c. Line Printer

d. CRT Display and Keyboard

3.3.1.1 Digital cmputer. The Digital Computer shall minimally provide:

a. Word size of 16 bits minimum.

b. Minimum of 64 K bytes of mainframe memory expandable to at least 256 K bytes.

c. Hardware multiply and divide or microprogramming capabiltiy.

d. Real Time Clock.

e. Power Fail/Automatic Restart.

f. Vectored priority interrupts.

g. Memory parity check and error correction.

h. One (1) spare Direct Memory Access (DMA) and one (1) spare I/O channel in addition to those channels used to implement this specification.

i. Computer execution times and access times as required to meet software processing requirements.

j. Read Only Memory (ROM) bootstrap loader.

k. Compatibility with the peripheral equipment specified in 3.3.1.2, 3.3.1.3, 3.3.1.4, 3.3.1.5 and 3.3.2.

l. Non-operating altitude: 15,000 feet maximum continuous.
30,000 feet maximum up to 12 hours.

m. Non-operating temperature: -20 C to +50 C.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the currentversion before use.

3.3.1.2 Disc drive(s). A dual magnetic disc drive unit (or two (2) individual disc drive units), shall contain two (2) discs at least one (1) of which is a removable disc cartridge. The drive unit(s) shall interface with the digital computer via a dedicated DMA channel. The disc drive shall minimally provide:

a. Average head positioning time of no more than 30 m sec.

b. Average rotational latency of no more than 10 m sec.

c. Data transfer rate of at least 7.5 megabits/second.

d. Storage capacity of 9.67 megabytes per disc.

e. Humidity

(1) Operating: 8 per cent to 80 per cent, non-condensing at wet bulb temperature to 25.6°C.

(2) Non-Operating: 5 per cent to 95 per cent, non-condensing at wet bulb temperature to 29.4°C.

3.3.1.3 Line printer. The line printer shall be of a dry-process type and shall minimally provide:

a. Print speed of 300 characters per second.

b. 72 American Standard Code for Information Interchange (ASCII) ANSI X3.4 characters per line. Proposed characters/line shall be consistent with display (Ref. 3.3.1.4 or 3.3.1.5) characters/line capability provided.

c. Printed copy of test results, test program listings, error messages and operator instructions.

d. Compatibility with fanfold paper.

e. Standard ASCII set of 64 characters.

3.3.1.4 Display/keyboard terminal. The display/keyboard terminal shall minimally provide:

a. A cathode-ray tube (CRT): Minimum 5" x 10" screen size.

b. Display capacity of 24 simultaneous lines and 72 characters/line.

c. Display data rate of 9600 baud.

d. Standard display set of 64 ASCII characters.

e. Standard ASCII character and control keyboard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

f. Display and keyboard controls shall be consistent with performance requirements herein specified.

g. Scrolling

h. Non-operating altitude: 25,000 feet, maximum.

i. Dual cartridge tape unit with up to 110 K bytes of storage on each cartridge tape.

3.3.1.5 Color graphics display terminal. Procurement requirements are identified in 6.2.1 contained herein for a Color Graphics Display Terminal minimally providing the features delineated below:

3.3.1.5.1 Display. The display type shall be a high resolution, 13 inch diagonal CRT with red, green and blue phosphors. Techniques to display data and data refresh rate shall be specified by the offeror.

3.3.1.5.2 Display characteristics. The display shall provide a minimum screen resolution of 512 x 256 picture elements (pixels). Each pixel will be addressable. It will be possible to display eight (8) colors simultaneously (including black and white).

3.3.1.5.2.1 Alphanumeric characters. It shall be possible to display a minimum of 24 rows with 72 characters per row. A standard display set of 64 ASCII characters shall be provided.

3.3.1.5.3 Keyboard. As a minimum, a standard ASCII character and control keyboard shall be provided. Keyboard controls shall be consistent with performance requirements herein specified.

3.3.2 Peripheral equipment. Acquisition of the peripheral equipment delineated below for Software Development Station(s) and/or HTS Stations specified herein will be deferred. However, the software necessary to accommodate these peripherals, shall form a part of this procurement package and shall be so quoted by the offeror. Software Development Station(s) will consist of some or all of the peripheral equipment and control subsystem elements defined within this specification.

3.3.2.1 Card reader. The card reader shall minimally provide:

a. Readout of standard 80 column card.

b. Reading speed of 250 cards per minute.

c. Hopper capacity of 600 cards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.2.2 Magnetic tape unit. The Magnetic Tape Unit shall minimally provide:

a. 9-track drive.

b. Recording density of 800 characters per inch NRZI.

c. Tape speed of 45 inches per second.

d. Read after write capability.

e. 10.5 inch diameter reel size.

3.3.2.3 High speed line printer. The High Speed Line Printer shall minimally provide:

a. 132 columns per line.

b. 64-character ASCII set.

c. Printing speed of 500 lines per minute.

d. Compatibility with fanfold paper.

3.3.3 Stimulus requirements. The system shall provide the following programmable stimuli:

3.3.3.1 Waveform synthesizer. The Waveform synthesizer shall minimally provide:

a. Sine, square, ramp (positive and negative), and triangle waveforms.

b. Amplitude programmable from 20 mv to 10 V P-P nominal into 50 ohm load with the following accuracies:

(1) Sine

Frequency	Dedicated	Universal
$f \leq 100$ KHZ	$\pm 1.4\%$ of full range, ≥ 3 Vp-p	$\pm 2.0\%$ of full range, ≥ 3 Vp-p
	$\pm 2.6\%$ of full range, < 3 Vp-p	$\pm 3.1\%$ of full range, < 3 Vp-p
100 KHZ $< f \leq$ 10 MHZ	$\pm 7.7\%$ of full range, ≥ 3 Vp-p	$\pm 8.3\%$ of full range, ≥ 3 Vp-p
	$\pm 10.1\%$ of full range, < 3 Vp-p	$\pm 10.7\%$ of full range, < 3 Vp-p

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

(1) Sine (Contd)

Frequency	Dedicated	Universal
10 MHZ<f≤20 MHZ	±17.6% of full range, ≥3 Vp-p	±18.8% of full range, ≥3 Vp-p
	±24.3% of full range, <3 Vp-p	±25.5% of full range, <3 Vp-p

(2) Square

Frequency	Dedicated	Universal
f≤100 KHZ	±11.4% of full range, ≥3 Vp-p	±11.9% of full range, ≥3 Vp-p
	±13.9% of full range, <3 Vp-p	±14.4% of full range, <3 Vp-p
100 KHZ<f≤10 MHZ	±14.1% of full range, ≥3 Vp-p	±14.7% of full range, ≥3 Vp-p
	±16.6% of full range, <3 Vp-p	±17.2% of full range, <3 Vp-p

(3) Triangle and Ramp

Frequency	Dedicated	Universal
f≤2 KHZ	±1.3% of full range, ≥3 Vp-p	±1.8% of full range, ≥3 Vp-p
	±2.5% of full range, <3 Vp-p	±3.1% of full range, <3 Vp-p
2 KHZ<f≤10 KHZ	±6.4% of full range, ≥3 Vp-p	±7.0% of full range, ≥3 Vp-p
	±7.6% of full range, <3 Vp-p	±8.2% of full range, <3 Vp-p

C. Amplitude resolution of 0.1% of full range over at least three (3) ranges covering the full amplitude capability.

d. Offset voltage programmability -4.5 V to +4.5 V on highest range, decreasing to ±4.5 mv on lowest range with degradation of +0.3% of full range for dedicated path and ±0.9% of full range for universal path.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

e. Source impedance of 50 ohms.

f. Frequency programmable through the following ranges:

(1)	SINE	0.01 HZ to 20.999 MHZ
(2)	SQUARE	0.01 HZ to 10.999 MHZ
(3)	RAMP	0.01 HZ to 10.999 KHZ
(4)	TRIANGLE	0.01 HZ to 10.999 KHZ

g. Frequency accuracy of $\pm$0.0005% of setting, 20° to 30°C.

h. Frequency resolution of 1 uHZ for f <100 KHZ and 1 mHZ for f $\geq$100 KHZ.

i. Frequency stability of $<1 \times 10^{-6}$ of frequency per month, 20° to 30°C.

j. Phase lock, continuous modes.

k. Sine wave amplitude modulation (available only at user interface):

(1) Modulation depth at full output for each range: 0-100%

(2) Modulation frequency range: DC-50 KHZ (0-20.999 MHZ carrier frequency)

(3) Sensitivity: $\pm$ 5V peak for 100% modulation

l. Sine wave phase modulation (available only at user interface):

(1) Range: $\pm$850°, $\pm$5V input

(2) Modulation frequency range: DC-5 KHZ

m. Distortion, linearity, slew rate, rise and fall time limitations shall be specified by the offeror.

3.3.3.2 Arbitrary waveform generator. Three (3) independent arbitrary waveform generator outputs which can be phase related shall be minimally provided. Each output shall be controllable by at least one (1) trigger input and one (1) cursor output. At

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

least one (1) sync output shall be provided for each arbitrary waveform generator output. Each arbitrary waveform generator output shall minimally provide:

a. Number of samples per waveform: 2 - 1024

b. Time of one (1) sample (stepping time from data point to next adjacent point):

(1)	Range:	200 nsec - 999.0 sec (fixed waveform) 500 nsec - 999.9 sec (arbitrary waveform)
(2)	Resolution:	100 nsec
(3)	Accuracy:	$\pm$0.1% of programmed value

c. Amplitude (into 50 ohm load, output impedance <1 ohm, fixed waveform):

(1)	Range:	8 mv - 20 VPP
(2)	Resolution:	3 digits
(3)	Accuracy (Dedicated Path):	$\pm$ (2.4% of programmed value $\pm$ 2.5 x least significant place value)
(4)	Accuracy (Universal path):	$\pm$ (2.8% of programmed value $\pm$ 2.5 x least significant place value)

d. Offset:

(1)	Range:	-10 to +10 VDC
(2)	Resolution:	3 digits
(3)	Accuracy (Dedicated Path):	$\pm$ (2.4% of programmed value $\pm$ 2.5 x least significant place value)
(4)	Accuracy (Universal Path):	$\pm$ (2.8% of programmed value $\pm$ 2.5 x least significant place value)

e. Slew rate: 80 V/u sec

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

f. Phase shift between any two (2) outputs:

(1)	Range:	200 nsec - 999.9 sec (fixed waveform)
		500 nsec - 999.9 sec (arbitrary waveform)
(2)	Resolution:	100 nsec
(3)	Accuracy:	±25 nsec ±0.1% of programmed value

g. Sine, Square, Ramp and Triangle Waveforms: Programming of "samples" to achieve these standard waveforms shall be provided by the HTS. Linearity and distortion versus frequency shall be specified for all waveforms by the offeror.

h. Sync Output, Trigger Input, and Cursor Output shall be Transistor Transistor Logic (TTL) compatible.

i. Non-operating altitude: 40,000 feet maximum.

3.3.3.3 Pulse generators. Each of two (2) Pulse Generators shall have as a minimum, a main output, a trigger input and a sync output. Each shall minimally provide (20° to 30°C):

a. A PRF programmable from 0.5 HZ to 25 MHZ. Resolution of 3 digits of programmed value. Accuracy of ±2% ±1 nsec.

b. Pulse delay shall be defined as the time delay between the rising edge of the sync output and the 0% point of the leading edge of the pulse for a single pulse output. For a double pulse output, pulse delay shall be defined as the time delay between the 0% point of the leading edge of the two pulses. Pulse delay for both single and double pulse outputs shall be programmable from 0 nsec and 3 digits of programmed value from 20 usec to 999 msec. Accuracy of ±1% ±10 nsec.

c. Pulse width shall be defined as the time between the 50% point of the rising and falling edges of the pulse. Pulse width shall be programmable from 10 nsec to 999 msec. Resolution of 1 nsec from 10 nsec to 19.999 usec and 3 digits of programmed value from 20 usec to 999 msec. Accuracy of ±1% ±10 nsec.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

d. Amplitude programmable from ±40 mv to ±10 V nominal into 50 ohm load from a 50 ohm source. Resolution of 25 mv for the full range. Amplitude or dc offset accuracy of:

(1)	Dedicated path (25 MHZ)	±9.2% of programmed value ±63 mv
(2)	Universal path (25 MHZ)	±11.5% of programmed value ±63 mv
(3)	Dedicated path (20 MHZ)	±7.1% of programmed value ±63 mv
(4)	Universal path (20 MHZ)	±8.3% of programmed value ±63 mv

e. Rise and fall time programmable from 10 nsec. to 25 msec. Accuracy of +5% ±10 ns over first 10:1 range. Linearity of less than ±7% for rise and fall time greater than 50 nsec (over first 10:1 range).

f. Sync output. Greater than 1.5 V into 50 ohms.

g. Jitter less than 0.1% of setting for frequency, pulse delay and pulse width.

h. Duty cycle 90% or greater.

i. Normal and inverted, single and double pulse modes.

j. Pulse distortion, settling time, overshoot, undershoot, duty cycle, droop and ringing limitations shall be specified by the offeror.

k. Non-operating altitude: 40,000 feet maximum.

3.3.3.4 Synchro/resolver. The programmable synchro/resolver simulator shall minimally provide synchro or resolver outputs with the following capabilities:

a. Programmability of synchro or resolver simulator function.

b. Angle range of 0° to 360°

c. Resolution of 0.02^{o}

d. Accuracy of $\pm 0.05^{o}$

e. Output signals programmable to 11.8V/26V/90V line to line @ 47 HZ to 1 KHZ.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

f. Reference inputs of 115V or 26V RMS @ 47 HZ to 1 KHZ.

g. Non-operating altitude: 40,000 feet maximum.

3.3.3.5 Digital unit. Digital Unit shall be Digital to the offeror as Government Furnished Equipment (GFE). A DWG Interface and Performance Specification, MIL-D-85424, which defines DWG system capabilities and integration requirements, is available to augment this procurement package.

a. The DWG Interface and Performance Specification details the specific interface requirements for integrating the DWG within an ATE system. Offeror shall utilize this specification to specify DWG characteristics at the ATE/UUT system interface of the HTS. Offeror's proposal shall define all hardware and software elements which are required to incorporate the DWG within the offeror's proposed HTS including, but not limited to, the following requirement: Input power to the DWG, DWG interface to control computer and DWG interface to the ATE/UUT interface.

b. Also provided as GFE will be a DWG Self-Test Program in Diagnostic Flow Chart (DFC) Format to be coded in ATLAS and implemented by the offeror within the proposed System Self-Test.

c. It is a requirement that the DWG external DC Reference requirements will be fully met by appropriate offeror utilization of the nine (9) Programmable DC Power Supplies and the four (4) Precision DC References cited in 3.3.3.7 and 3.3.5.1.1., respectively.

3.3.3.6 Manchester capability. The HTS Manchester Interface shall be capable of supporting equipment which has been designed in accordance with the requirements of MIL-STD-1553. The Manchester Interface shall provide four basic modes of operation:

a. Transmit. In this mode, the interface is programmed to transmit up to 1K words. The receive portion of the interface is not operational during this mode.

b. Receive. In this mode, the interface is programmed to receive up to 1K words. The transmit portion of the interface is not operational during this mode.

c. Transmit/receive. In this mode, the interface is programmed to transmit a fixed amount of data to a UUT and, upon completion of the transmission, enable reception of a fixed amount of data from the UUT.

d. Receive/transmit. In this mode, the interface is programmed to receive fixed amount of data or detect a particular pattern from a UUT and, after a programmed delay, transmit a fixed amount of data to the UUT.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.3.6.1 Programmable bidirectional data. The interface shall provide two programmable bidirectional data channels and one set of the following output lines per channel:

a. Output data clock. This clock shall be that clock that is used by the interface to encode Manchester data (transmitter).

b. A sync signal shall provide a gating signal for external equipment to indicate channel activity (transmitter).

c. Recovered input data clock. This clock shall be derived from the input data (self-clocking). This derived input data clock shall be the "Recovered Input Data Clock" (received).

3.3.3.6.2 Clock characteristics. The "Output Data Clock" and the "Recovered Input Data Clock" shall be compatible with the Digital Word Generator's (DWG's) "External Clock Input" requirements thereby providing the capability to synchronize the DWG to Manchester data.

3.3.3.6.3 Data channel parameters. The following parameters shall be independently programmable for each data channel:

a. Transmission bit rate: 0.125 to 2.0 megabits/second. Bit rate accuracy of $\pm$0.1%.

b. Bit time: 0.5 to 8.0 microseconds in 0.250 microsecond increments.

c. Bit transistion time: The bit transition time shall be 0.250 to 6.0 microseconds in 0.125 microsecond increments. The MIN/MAX range formula for bit transition time is as follows:

0.250 (Bit Time) $\leq$ Bit Transition time $\leq$0.75 (Bit Time) for bit time $\geq$1.0 usec

0.250 usec $\leq$ Bit Transition Time $\leq$(bit time -0.25 usec) for bit time $<$ 1.0 usec

d. Word size: 14 to 24 bits (includes parity but not sync).

e. Sync characteristics. The Manchester Interface shall provide programmable Sync Characteristics for transmission and reception as described below:

(1) Transmitter single transition characteristics: Sync time shall be programmable from 1 to 24 microseconds

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

in 0.250 microsecond increments. Sync transition time shall be programmable from 0.250 to 18 microseconds in 0.125 microsecond increments. The MIN/MAX range formula for sync transition time shall be defined as:

0.25 (Sync time) $\leq$Sync transition time $\leq$0.75 (Sync time)

(2) Transmitter multiple transition characteristics: Number of sync cells shall be programmable as either four or six cells. Sync celltime duration shall be programmable from 0.25 microsecond to 4 microseconds in 0.25 microsecond increments (all cells have equal time duration). Sync cell output level shall be programmable to provide either a positive or negative line to line output voltage (capability of providing a transition at each intercell boundary. Total of one transition minimum to five transitions maximum).

(3) Receiver sync characteristics: Sycn time shall be programmable from 1 to 24 microseconds in 0.25 microsecond increments. Sync transition time shall be programmable from 0.25 microsecond to 18 microseconds in 0.125 microsecond increments. The MIN/MAX range formula for sync transition time shall be defined as:

0.25 (Sync time) $\leq$ Sync transition time $\leq$ 0.75 (Sync time).

f. Parity: Odd, even or none.

g. Dead time: 0 to 20 usec in 0.125 usec increments.

i. Receiver input level (Threshold): 1.0 to 10 volts p-p in 0.1 volt increments with receiver input impedance: 35 ohms $\pm$5 ohms.

j. Delay offset between programmable channels: 0 to 50 microseconds in 0.25 usec increments.

h. Insertion of invalid Manchester II code on any or all bits of a word.

3.3.3.6.4 Manchester interface. The HTS Manchester Interface shall provide the following.

3.3.3.6.4.1 Data transmission/reception. The interface shall be capable of transmitting or receiving up to 1 K words contiguously, per channel, or 2 K words contiguously if only one channel is used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.3.6.4.2 Remote terminal (UUT) response times. The interface shall have the capability to accept remote terminal response times as specified in MIL-STD-1553 and as described below.

a. Terminal fail-safe. When the Manchester channel is in the transmit/receive mode and the UUT does not respond in the programmed amount of time, an error condition shall be detected, the reception aborted, and a "Done" indication sent to the processor.

b. Channel response time. When the Manchester channel is being used as a remote terminal to test a controller, the response time in this case shall be the amount of delay before a word is transmitted to the UUT.

3.3.3.6.4.3 Data bus characteristics. The Manchester Interface shall have the capability to interface with equipment designed in accordance with MIL-STD-1533.

3.3.3.6.4.4 Data characteristics. The Manchester Interface shall have Data Characteristics in accordance with MIL-STD-1553 as specified below:

a. Data code.

b. Word formats.

c. Message formats.

3.3.3.6.4.5 Pattern match. The Manchester Interface shall have the capability to be programmed to detect a particular pattern, and upon detection of the pattern perform the following operations:

a. Step to any one of the four basic modes of operation (reference 3.3.3.6).

b. Notify processor that the particular pattern specified has been detected.

3.3.3.7 Programmable DC power supplies. The DC Power Supplies shall minimally provide the following capabilities:

a. Remote sense

b. Over-voltage protection

c. Current limiting

d. Series connection

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

e. Line Regulation of 0.05% for any line voltage change within rating

f. Resolution of 0.05% of full scale

3.3.3.7.1 DC power supplies minimum requirements.

Quantity	Range VDC, Max. Current	Tolerance	Ripple
3	0 to 36V, 3A	1%	5 mv max.
1	0 to 25V, 4A	1%	5 mv max.
1	0 to 15V, 6A	1%	5 mv max.
1	0 to 55V, 2A	1%	5 mv max.
1	0 to 6V, 10A	1%	5 mv max.
1	0 to 100V, 1A	1%	10 mv max.
1	0 to 200V, 1A	1%	10 mv max.

3.3.3.7.2 Load regulation. The power supplies shall have the following minimum load regulation of no load to full load at full voltage on dedicated pins:

a. With remote sense.

(1)	0 to 6V, 10A	0.05% 1/
(2)	0 to 25V, 4A	0.05%
(3)	0 to 15V, 6A	0.05%

b. Without remote sense.

(1)	0 to 36V, 3A	2.3% 2/
(2)	0 to 55V, 2A	1.4%
(3)	0 to 100V, 1A	0.33%
(4)	0 to 200V, 1A	0.06%

1/ When load current exceeds 7 amp, the two interface pins for each output shall be connected.

2/ Maximum current supplied by 0 to 36V, 3A power supplies is 2 amp using universal pins.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.3.8 AC power supply. The AC power supply shall minimally provide:

a. 3-phase, 4-wire output

b. Independent amplitude programming of each phase

c. 167 volt-amperes per phase, 0.5 to 130 V RMS outputs (line to neutral), power factor ±0.7 to 1

d. Tap on each phase to provide 26 V RMS when the phase has been programmed to 115 V RMS

e. Amplitude accuracy of ±1.5% of full scale under all conditions (includes load regulation), 0.1V RMS resolution

f. 45 HZ to 5 KHZ output frequency range

g. ±1.3% of programmed value frequency accuracy

h. 1 HZ resolution through 999 HZ, 10 HZ resolution for frequencies greater than 999 HZ

i. Output distortion of less than 1%

j. Outputs isolated from signal and chassis ground

k. Overload and short circuit protection

3.3.4 Response requirements. The system shall provide the following programmable measurement capability:

3.3.4.1 Digital multi-function meter. The digital multi-function meter shall minimally provide:

a. Measurements to a precision of at least 5 1/2 digits

b. 4 ranges covering the full amplitude capability. Ranging shall be automatic for all functions.

c. True RMS AC measurement capability

d. Resistance measurements of 0 to 10 M ohms to the following accuracies (with auto-calibration on):

(1) 4-wire measurement (dedicated pins)

0.1 K ohm range:	±(0.0074% x reading + 0.008 ohms)
1 K ohm range:	±(0.0074% x reading + 0.012 ohms)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

(1) (Contd)

10 K ohm range: ±(0.0094% x reading + 0.216 ohms)

100 K ohm range: ±(0.0064% x reading + 3.16 ohms)

1000 K ohm range ±(0.018% x reading + 51.6 ohms)

10,000 K ohm range: ±(0.132% x reading + 516 ohms)

(2) 4-wire measurement (universal pins)
Same as 4-wire measurement on dedicated pins

(3) 2-wire measurement (dedicated pins)
Same as 4-wire measurement on dedicated pins except add 2.5 ohms

(4) 2-wire measurement (universal pins)
Same as 4-wire measurement on dedicated pins except add 3.9 ohms

e. DC Voltage Measurements (with auto-calibration on): @ ±(0.0124% x Rdg. + 0.0011% x range) accuracy to 200 V with input impedance ≥10 M ohms for all DC ranges.

f. AC Impedance: Input resistance of greater than 1 M ohms; shunt capacitance of less than 350 pf.

g. AC Voltage Measurements:

(1) @ ±(0.06% Rdg. + 0.06% range) accuracy to 200 V at frequencies to 20 KHZ

(2) @ ±(0.616% Rdg. + 0.178% range) accuracy to 200 V at frequencies to 100 KHZ

(3) @ ±(2.116% Rdg. +0.348% range) accuracy to 10 V at frequencies to 250 KHZ

(4) @ ±(5.116% Rdg. + 0.648% range) accuracy to 10 V at frequencies to 500 KHZ. 1/

(5) @ ±(6.316% Rdg. + 3.548% range) accuracy to 10 V at frequencies to 1 MHZ. 1/

1/ Measurements above 250 KHZ may be made via waveform digitizer (see 3.3.4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.4.2 Counter-timer. The dual channel Counter-Timer shall minimally provide:

a. Measurements to a precision of at least 8 1/2 digits.

b. Programmable capability to measure frequency, period, period average, time interval, time interval average, rise time, fall time, pulse width and events.

c. Automatic trigger settings for frequencies $\geq$ 400 HZ and voltages $\geq$ 1 VRMS.

d. Automatic rise, fall time, and pulse width measurements.

e. Frequency measurements from 0 to 100 MHZ (instrument spec). Frequency measurement capability from 100 MHZ to 512 MHZ available on a third channel under local mode of operation. Accuracy for frequency measurements from 0 to 100 MHZ shall be:

$\pm$1 count $\pm$ reference error

where

$$1 \text{ count} = \frac{10^{-8}}{\text{gate time, sec}} \times \text{input frequency (HZ)}$$

$$\text{reference error} = f_r \times a \times t$$

and

f_r = reference frequency of internal oscillator, HZ

a = aging rate = 3×10^{-7}/month

t = elapsed time since instrument has been calibrated off-line, months

f. Period measurements from 10 nsec to 10^9 seconds with an accuracy of:

$\pm$1 count $\pm$ reference error $\pm$ trigger error

where

$1 \text{ count} = 10^{-8}$ second

$$\text{reference error} = Eo + \left(\frac{fo}{fr} - 1 \right) t + \frac{at^2}{2}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

and

Eo = initial time error after calibration off-line, sec

f_o = initial frequency of internal oscillator after calibration off-line, HZ

f_r = reference frequency of internal oscillator, HZ

t = elapsed time since instrument has been calibrated off-line, months

a = aging rate = 3×10^{-7}/month

$$\text{trigger error} = \frac{0.3}{(S/N) \times \text{input frequency}}, \text{ sec (sine wave)}$$

and

S/N = Signal-to-noise ratio of input, V.

g. Period average measurements from 10 nsec to 1 second with an accuracy of:

$$\pm \text{ count} \pm \text{reference error} \pm \frac{\text{trigger error}}{\text{number of periods averaged}}$$

Definition of each term shall be the same as period measurement.

h. Time interval measurements from 10 nsec to 10^9 seconds with an accuracy of:

$$\pm \text{ count} \pm \text{reference error} \pm \text{trigger error}$$

where

1 count = same definition as period measurement

reference error = same definition as period measurement

$$\text{trigger error} = \frac{0.0025 \text{ usec}}{\text{signal slope (V/usec)}}$$

and signal slope shall be the smaller of the two signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

i. Time interval average measurements from 10 nsec to 1 second with an accuracy of:

$$\pm \text{ reference error } \pm 2 \times 10^{-9} \pm \frac{(\text{trigger error} + 10^{-8})}{\sqrt{(\text{Number of intervals averaged})}}$$

where

reference error = same definition as period measurement

trigger error = same definition as time interval measurement

j. Software start/stop

k. Input impedance of 1M or 50 ohms programmable

l. Input signal ranges to $\pm$ 200V

m. Input trigger levels programmable to $\pm$ 200 V in three ranges with an accuracy of $\pm$5% of range. Resolution of three (3) digits.

n. Minimum sensitivity of 100 millivolts RMS at 50 MHZ, sine wave.

o. AC and DC coupling.

p. Internal clock stability - Temperature stability $< 5 \times 10^{-6}$ of frequency from $+10^{o}$C to $+40^{o}$C.

3.3.4.3 Waveform digitizer. The Waveform Digitizer shall be a two (2) channel instrument capable of digitizing and storing sampled values of analog waveforms from DC to 25 MHZ minimum. Software shall be provided to utilize this data to produce the following standard waveform analysis measurements using ATLAS:

a. Peak volts

b. Peak to peak volts

c. Average volts

d. RMS volts

e. Offset

f. PRF

g. Pulse Width

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the currentversion before use.

h. Phase Angle

i. Distortion

j. Rise/Fall Time

k. Spectrum Analysis

Capability shall also be provided to transmit all digitized data to the computer for other programmed measurements.

3.3.4.3.1 Waveform digitizer detailed specification. The Waveform Digitizer shall minimally provide:

a. 2K Bytes sampled data storage memory.

b. Programmable transient or delayed sampling modes.

c. Sampling interval in normal or delayed modes programmable from 10 nsec. to 1 sec.

d. Two (2) independent channels which may be alternately sampled.

e. AC or DC coupling.

f. Internal or external triggering.

g. Delayed triggering between two channels of up to 2048 sample intervals.

h. A/D resolution of 8 bits.

i. Input signal range of ±50 mv to ±200 V in at least 8 programmable ranges (i.e. 10 db steps or less).

j. Input Impedance: Programmable - high/low
High: Input Resistance of 1 Mohm; shunt capacitance of 25 pf (at instrument)
Low: 50 ohms

k. Measurement bandwidth of DC to 25 MHZ.

l. Maximum signal power of 20 watts average (when) terminated in 50 ohms).

m. Measurement accuracy into 50 ohms of the following representing the combined hardware accuracies of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

programmable attenuator and sampling system (accuracies due to software measurement algorithms shall be separately specified):

(1) f <300 KHZ
Dedicated: ±2.3% of reading
Universal: ±2.8% of reading

(2) 300 KHZ < f <20 MHZ
Dedicated: ±7.7% of reading
Universal: ±8.9% of reading

3.3.4.4 Synchro/resolver indicator. The programmable synchro resolver indicator shall minimally provide position measurements on one (1) synchro or resolver input with the following capabilities:

a. Programmability of synchro or resolver indicator function.

b. Angle range of 0° to 360°.

c. Resolution of 0.02^{o}.

d. Accuracy of $\pm 0.03^{o}$.

e. Input signals of 11.8V/26V/90V line to line, 47 HZ to 1 KHZ.

f. Reference input range of 10V to 130V RMS, 47 HZ to 1 KHZ.

g. Non-operating altitude: 40,000 feet maximum.

3.3.4.5 Probe capability. The HTS shall incorporate a probe or probes which shall provide programmable access to any unique response input of any HTS analog or digital measurement resource. The probe interface shall minimally meet system response requirements cited herein.

3.3.4.5.1 Guided probe. A Guided Probe shall be provided for use by the HTS operator under direction of a test program. The characteristics of this probe and its interfaces to the HTS shall be such that the digital measurements necessitated by ATG software described in 3.3.7.3.1 can be accomplished. The probe shall be capable of readily penetrating UUT conformal coating.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the currentversion before use.

3.3.5 Interface/switch characteristics.

3.3.5.1 Interface pin requirements. The interface shall minimally provide a total of 600 pins. An estimated pin allocation consistent with requirements specified herein is:

Type of Signal	Quantity
Universal (Switching) Pins	120
Non-Universal Pins:	
High Performance (coaxial)	44
Dedicated and/or Functional (solid)	47
Digital Word Generator Pins:	
Stimulus/Response	240
External Sync	12
External DC Ref	6
Grounds	10
Manchester Pins (2 channels)	14
Contact Closures - Signal	48
Grounds	6
Spares	53
TOTAL:	600

An additional seven-pin connector for the ac power supply shall be provided.

3.3.5.1.1 Universal switching. Universal switching is defined herein as the capability to simultaneously switch (provide) any or all analog stimulus or response instrument functional input/output lines to any or all universal interface pins.

Analog stimulus as well as response lines which are not switched universally (as the term is defined above), shall be provided via dedicated or functional switching such that system performance flexibility shall not be unreasonably compromised. All changes to quantities and types of signal switching capabilities defined in 3.3.5.1 shall be fully described by the offeror.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

a. Universal switching capability proposed by the offeror shall be clearly defined. Such definition should address:

(1) Analog stimulus or response lines not universally switched.

(2) Dedicated and functional switching provided in lieu of universal switching.

(3) Number of unique simultaneous switch closures which are possible.

(4) Sparing provisions for increasing the number of analog stimulus/response instrument lines which can be universally switched.

3.3.5.1.1.1 Universal switch path characteristics. The characteristics of the universal switch paths shall be:

a. Leakage Resistance: >100 M ohms to ground

b. Bandwidth (3 DB): ≥50 MHZ

c. Crosstalk (between adjacent universal pins):

-40 DB or better, DC to 1 MHZ
-35 DB or better, DC to 10 MHZ

d. Shunt Capacitance (to ground): ≤500 pf

e. Maximum Voltage: Not less than 200 V

f. Maximum Current: ≥ 2 amps, steady state
≥ 1 amp switched at 28 VDC (resistive load)

g. Nominal Path Resistance Through the Switch: ≤0.5 ohms

h. Terminations: Selectable terminations at any universal pin.

(1) 50 ohms ±5%, 2 watts

(2) 100 ohms ±5%, 1 watt

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.5.1.1.2 Precision DC references. Four (4) precision DC reference sources shall be available at all universal pins and shall provide the following as a minimum:

a.	Voltage:	-10 V to +10 V
b.	Voltage Accuracy:	±1%
c.	Voltage Resolution:	0.5% of full scale
d.	Maximum Ripple:	≤5 mv
e.	Maximum Current:	Not less than 100 ma

3.3.5.1.2 High performance analog. As a minimum, one (1) high performance dedicated path shall be provided for each analog instrument functional input/output line so as to minimize its associated path degradation. Dedicated switch paths shall be provided for all high performance analog instrument functional input/output lines which are also universally switched.

a. The following characteristics shall minimally be provided for High Performance Analog Instruments:

(1) Leakage Resistance: >100 M ohms to ground

(2) Bandwidth (3 DB): ≥70MHZ

(3) Crosstalk (between adjacent high performance pins)

-60 DB or better, DC to 1 MHZ
-45 DB or better, DC to 10 MHZ

(4) Shunt Capacitance (to ground): ≤ 350 pf

(5) Maximum Voltage: Not less than 200 V

(6) Maximum Current 1/

≥ 2 amps, steady state

≥ 1 amp, switched at 28 VDC (resistive load)

1/ Dedicated/Switched Power Supply output lines shall be provided to accommodate power supply current requirements exceeding the offeror's universal switch ratings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.5.1.2.1 High Performance analog allocation. An estimated composition (allocation and quantities) of high performance lines is provided herein. However, the actual composition, which is dependent upon the offeror's specific implementation of HTS functional requirements, shall be defined by the offeror.

a. Estimated Composition - 98 Pins Allocated

	Instrument	High Performance I/O Line
(1)	Digital Multi-Meter	High, RS+ Low, RS- Guard
(2)	Counter Timer	Chan A Chan B Ext. Gate Control Gate out
(3)	Waveform Digitizer	Chan A Chan B Ext. Trigger A Ext. Trigger B
(4)	Synchro/Resolver Indicator	S1/R1 S2/R2 S3/R3 R4 Ref High Ref Low
(5)	Synchro/Resolver Simulator	S1/R1 S2/R2 S3/R3 R4 Ref 1 High Ref 2 High Ref Low
(6)	AC Power Supply 1/	Ø A Ø A tap Ø B Ø B tap Ø C Ø C tap Neutral

1/Dedicated/Switched Power Supply output lines shall be provided to accommodate power supply current requirements exceeding the offeror's universal switch ratings.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

	Instrument		High Performance I/O Line
(7)	DC Power Supplies 1/ 0-36 V, 3A	#1	High (2) Low (2)
(8)	0-36 V, 3A	#2	High (2) Low (2)
(9)	0-36 V, 3A	#3	High (2) Low (2)
(10)	0-25 V, 4A	#4	High (2), RS+ Low (2), RS-
(11)	0-15 V, 6A	#5	High (2), RS+ Low (2),RS-
(12)	0-55 V, 2A	#6	High (2) Low (2)
(13)	0-6 V, 10A	#7	High (2), RS+ Low (2), RS-
(14)	0-100 V, 1A	#8	High (2) Low (2)
(15)	0-200 V, 1A	#9	High (2) Low (2)
(16)	Pulse Generator (2 required)		Pulse Generator #1 (2) Pulse Generator #2 (1) Trig In (2) Sync Out (2)
(17)	Waveform Synthesizer		Output Amp Mod Input Phase Mod Input
(18)	Arbitrary Waveform Generator (3 channels required)		Signal Output (3) Input Trigger (3) Cursor Output (3) Sync Output (3)
(19)	Probe		One (1)

1/ The full capabilities of the AC Power Supply and the DC Power Supplies as specified in 3.3.3.8 and 3.3.3.7 respectively shall be available on these pins.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.5.1.2.2 Universal allocation. The quantity of universal lines allocated to each instrument is provided herein (Figure 1).

3.3.5.1.3 Dedicated and functional switching. Dedicated and functional switching shall be provided as necessary to meet requirements specified in 3.3.5.1.1. Unused dedicated and functional pins may be provided as spare pins.

3.3.5.1.4 Digital word generator pins.

3.3.5.1.4.1 Stimulus/response. These 240 pins shall be the unswitced DWG interface pins referenced in 3.3.3.5.

3.3.5.1.4.2 External sync. These twelve (12) pins shall be the unswitched DWG external sync pins referenced 3.3.3.5.

3.3.5.1.4.3 External DC references. These six (6) pins shall be the unswitched DWG external DC Reference inputs referenced in 3.3.3.6.

3.3.5.1.5 Manchester pins. These fourteen (14) pins shall be the unswitched Manchester input/output lines referenced in 3.3.3.6.

3.3.5.1.6 Contact closures - signal. A minimum of sixteen (16) independently programmable Form C (SPDT) relays shall each provide three (3) utility pins at the system interface for miscellaneous signal switching. The characteristics of these switching elements shall be as follows:

a. Maximum Voltage: Not less than 200 V

b. Maximum Current: $\geq$2 amps, steady state

$\geq$1 amp, switched at 28 VDC (resistive load)

3.3.5.1.7 Grounds. A minimum of six (6) system ground pins shall be available at the system interface for the purpose of returning DC power and signal current. Each pin shall be rated to carry a minimum of ten (10) amps.

3.3.5.1.8 Spares. The interface shall include approximately fifty-three (53) spare pins for expansion capability.

3.3.6 Hardware documentation. Hardware documentation shall be provided and such data shall consist of the following items:

a. Source control drawings for off-the shelf hardware.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

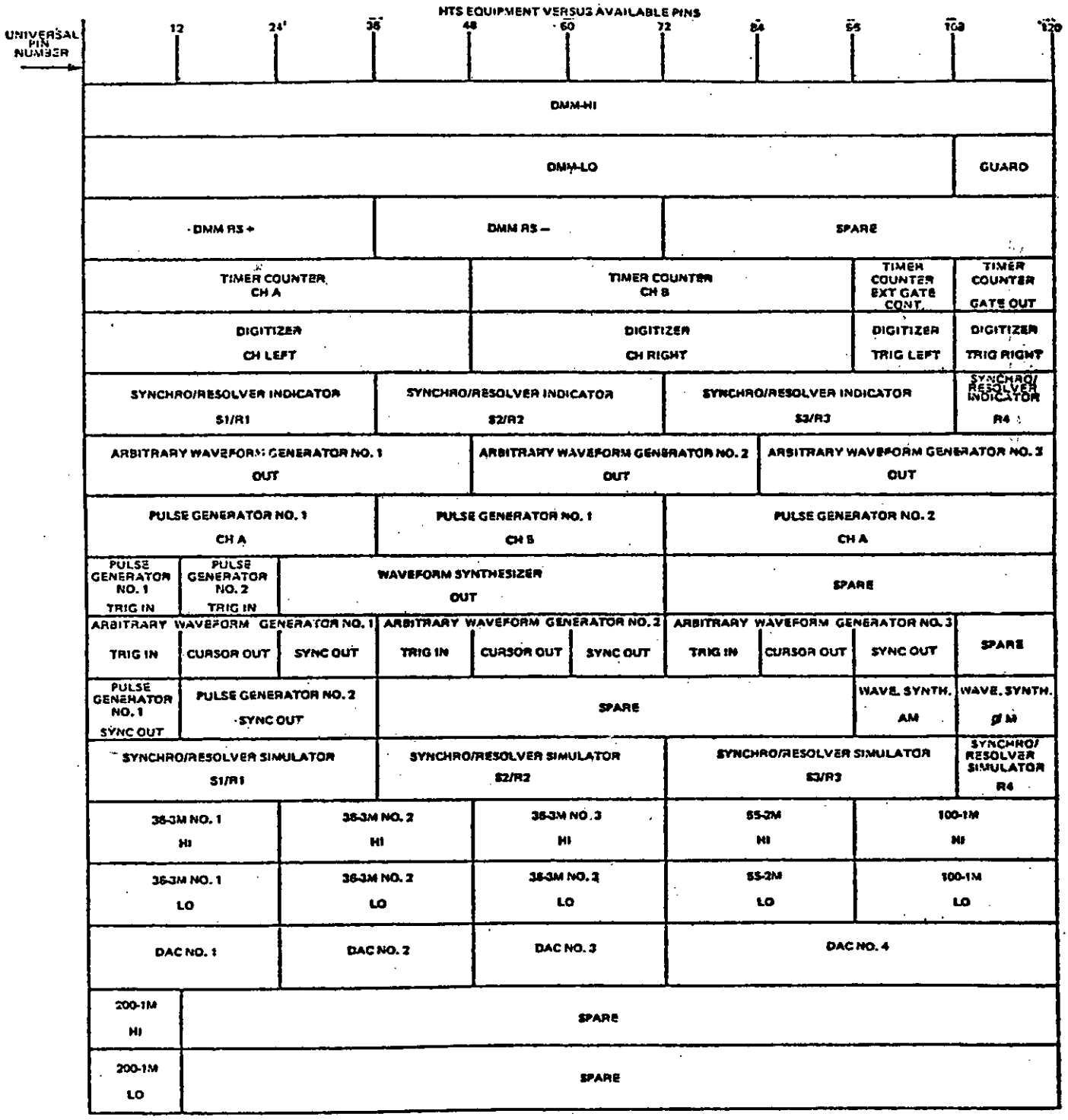


FIGURE 1. Universal pin switching.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:50Z
Check the source to verify that this is the current version before use.

b. System interconnection diagram

c. System arrangement drawings

d. Rack manufacturing drawing

e. Manufacturing drawings for system build-up and assembly.

f. Manufacturing drawings for system interconnection cabling.

g. Manufacturing drawings for hardware requiring new development efforts and for self-test adapter.

h. System specification.

3.3.7 System software. The system software shall be inclusive of Control Software (Operating System), Language Processors, and Automatic Test Program Generator (ATPG) Processors.

a. System software, as defined herein, includes all software modules necessary for test station opertion and system software maintenance. The software shall be of modular architecture such that software modules required only for software maintenance (reconfiguration, rebuild, etc.) may be omitted from operational ATE stations.

b. Software development station(s) to be configured/ acquired at a later date, will consist of designated software modules defined herein and designated hardware elements referenced in 3.3.1 and 3.3.2.

3.3.7.1 Control software. The operating system shall feature priority scheduling, background and foreground partitioning and a file management system. Multi-user operation shall allow concurrent test program execution, preparation and editing at, as a minimum, one (1) master and two (2) slave terminals. Concurrent testing of two (2) or more UUTs shall not be required. The following capabilities shall minimally be included:

a. File Manager

b. Loader

c. Operator Communication

d. Peripheral I/O Handler

e. Spooling

f. Batch processing capability

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the current version before use.

g. Multi-tasking

h. Interactive Editor

i. Media Conversion

j. Utilities and Operator Aids

k. Maintenance Management Information System (MMIS)

3.3.7.1.1 File manager. The File Manager shall support the dynamic creation, read/write access, modification, and deletion of any files on disc. File read/write access may be in either sequential or random access modes. Test programs, however, will be stored on disc, identified by name and directory and stored in a contiguous file. Any given file may be protected by use of file locks (unique password for any file or files). The File Manager shall maintain a file directory and a file data directory function, and return error indicators to the using program when:

a. Nonexistent file access is attempted

b. Attempts are made to modify or delete a protected file

c. Other logically inconsistent operations are attempted

When the file manager is used to create, move or delete files (by user interactive control), the file manager shall communicate this data to the user via CRT display or line printer.

3.3.7.1.2 Loader. (References are to system software programs only. See 3.3.7.2.3 for Test Program loading via ATLAS Processor.) The loader shall transfer programs of arbitrary size into computer memory via overlay techniques. Input programs may be resident on disc, or on any other input utility or related communications interface. This operation shall only require that the user identify the device of origin, and the identity of the file to be loaded. The loader shall also assemble various relocatable segments of program coding into object form, and produce prompts to the operator to assist in this process. The loader output shall consist of a map of resulting memory locations, plus the object file resident at the designated destination. (The loader may be used to load programs into computer memory, or may also, as an operator option, be used to catalog the assembled program file on disc, or on any other output utility of suitable capabiltiy.)

3.3.7.1.3 Operator communications. The Operating System shall minimally provide the operator with the capability to:

a. Edit, when not protected by a file lock,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

execute or catalog a UUT test program written in ATLAS Language.

b. List or display a program, or any portion(s) thereof.

c. Use language processors (ATLAS, FORTRAN IV & Assembler).

d. Monitor current status of tasks queued and active; i.e. multiple edit.

e. Selectively initiate or cancel tasks in the system.

f. Execute any catalogued object module.

g. Perform batch processing.

3.3.7.1.4 Peripheral I/O handler. The Device I/O Handlers for the computer peripherals and instruments shall be interrupt driven handlers to effectively use the multiprogramming capabilities of the system.

3.3.7.1.5 Spooling. The Operating System shall provide a mechanism for queuing peripheral output data on the system mass-store. The minimum spooling capability to be provided shall enable the automatic queuing of data destined for the line printer.

3.3.7.1.6 Batch processing. In a multiple console configuration, test program execution shall be performed on a "master" console and shall have the highest priority. Subject to this constraint, the operating system shall provide the ability to submit system software task(s) (list of commands) to the batch processor for subsequent, automatic system initiation. The operator shall queue tasks via task priority assignments. Nested batch runs and multiple concurrent batch runs shall be possible. Suitable operator messages shall be produced as each batch-controlled task is initiated.

3.3.7.1.7 Interactive editor. The Interactive Editor shall operate at any station and shall provide the ability to modify any source-format information when not file protected. The file system shall be accessed through the software operating system. The Interactive Editor shall display alphanumeric information on the station CRT and shall support and permit the direct modification of test so displayed using the CRT editing features. Such modification shall be unrestricted and shall require additional operator directives to produce permanent modification to the alphanumeric information on the file system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

a. Specifically, the Editor shall minimally provide for:

(1) Changes to any portion of any line of coding.

(2) Adding new lines, deleting old ones, or changing the position of groups of lines.

(3) Displaying any source file or portion thereof, or listing any program or portion thereof to the printer.

(4) Editing one file by appending or inserting another.

(5) Performing any of the above via programmed editor instructions from another file as well as via keyboard instructions.

(6) Character string replacement or deletion.

3.3.7.1.8 Media conversion. The media conversion software shall provide the capability to transcribe files among the available media such that data integrity and identification are assured. The media conversion software shall provide the capability to acquire data resident on one type of media and to transcribe it to another type of media. Depending on the specific peripherals connected to the station, the media control software shall minimally implement the following transcription processes:

a. Punched cards to mass-store

b. Magnetic tape to mass-store (source & object)

c. Mass-store to magnetic tape (source & object)

d. Mass-store to hard-copy (listings, source & object)

e. Mass-store to mass-store.

3.3.7.1.8.1 File comparison. The system software shall provide a difference detection function which shall detect and identify specific differences in two (2) source files, both of which are on mass-store. A listing of detectable differences shall be produced.

3.3.7.1.9 Multi-Tasking. The capability shall be provided to process multiple tasks concurrently. This will allow UUT test program execution and editing of other test programs concurrently, subject to the limitation set forth in 3.3.7.1.6 namely that test program execution shall have the highest priority.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.7.1.10 Utilities and operator aids. System clear, single step, single test, test entry points, cycle test, and halt at test shall be minimally incorporated and initiated by operator keyboard entry.

a. The HTS shall provide the capability to display HTS resource status in response to an operator keyboard request. HTS resource status shall, as a minimum, include a list of the setup and configuration of all active stimulus, measurement, and switch resources at the time of operator request.

b. The software system shall provide a repertoire of utility subroutines which provide general mathematical functions inclusive of absolute values, exponentiation, integer values, common and natural logarithms, square roots, boolean solutions, conditional branching, trigonometric functions and processing functions such as conversion to and from floating point format, ASCII to octal, bit manipulations, I/O formatting, conversions to/from the internal code (hexadecimal, octal) and decimal modules.

3.3.7.1.11 Maintenance management information system (MMIS). Refer to 3.2.7.5.

3.3.7.2 Language processors (ATLAS, FORTRAN IV, assembler). The software shall provide the capability for interlanguage linking and variable sharing between all Test System Languages including ATLAS, FORTRAN IV, and Assembly Language. Any language herein identified shall be able to link to a program or a subroutine written in any other language. Provisions shall be incorporated to allow the sharing of files, data and program variables. Files created in one (1) language shall be usable by all other languages herein identified.

3.3.7.2.1 Assembler. The system software shall contain an Assembly Language Processor supporting the generation and maintenance of assembly level code for the station computer. The Assembly Language Processor shall input the file containing the assembly statement and shall develop object code in relocatable format. It shall be possible to produce a listing of the source and object code complete with symbol locations (relative to the beginning of the source program) and error flags/comments where appropriate.

3.3.7.2.2 Fortran IV. The system software shall provide a language processor for the compiling of Fortran IV source language. The source language accepted by the language processor shall be Fortran IV as specified in ANSI American National Standard Fortran X3.9-1978. The Fortran language processor shall read the file containing the Fortran IV statements and shall generate relocatable object code which, when processed through the loader, will be executable on the station computer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

It shall be possible to produce a listing of source and object code. The compiler shall provide syntax and semantic checking with identification of errors by line numbers.

3.3.7.2.3 ATLAS. The ATLAS processor shall be an interpretive on-line processor of the syntax and semantics language requirements of IEEE ATLAS Standard 416-1978. The ATLAS processor shall provide for a true subset of the ATLAS language adequate to facilitate full testing with the HTS. Adaptations or extensions to the current IEEE ATLAS are nonpreferred but will be considered if fully justified by the vendor and are supported by ATLAS proposals in IEEE ATLAS format for presentation to the IEEE ATLAS Committee for subsequent introduction into the ATLAS language.

a. The ATLAS processor structure shall be of a modular structure such that future extensions to the ATLAS language can be easily appended as necessary. The documentation, design, and structure of this processor shall be such that creation and/or modification of resource drivers is readily achieveable and implementable utilizing only the test system resources.

b. Design of the ATLAS processor shall be implemented in the most efficient manner possible consistent with the inclusion, at a minimum, of capabilities for:

(1) Access and translation of ATLAS source code to machine code from HTS disc or keyboard inputs.

(2) Editing of input statements to ensure proper syntax and semantics. This capability shall include rejection of improper statements and notifications of errors to the operator via diagnostic messages.

(3) Loading and executing correct ATLAS statements.

(4) Performing engineering checks on ATLAS statements to ensure station safety and guarantee that station limitations are not exceeded.

(5) Allow for processing of ATLAS statements such that one (1) ATLAS program may be executed without interference or delay while another is being assembled.

(6) Allow for full ATLAS edit capability, pursuant to 3.3.7.1.7 contained herein, without the requirement to recompile any other program statements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.3.7.3 Automatic test program generator processor(s). The HTS software shall contain one (1) or more preprocessors which as a minimum, shall translate digital Automatic Test Program Generator (ATPG) output data of SMC 3100, D4 LASAR output format into ATLAS source language programs which can be executed by the ATE.

3.3.7.3.1 Guided probe software. The HTS software shall contain a processor to implement a guided probe capability. This capability will optimize fault isolation resolution beyond that obtainable using UUT interface pins alone by using test patterns and probe placement data provided by the ATPG data for guided probe application.

3.3.7.4 Software documentation. For implementation and maintenance of, revision to, and configuration control of system software (generated to best commercial standards), software documentation shall be developed and shall consist of the following items:

a. Software Family Tree identifying all system software to the module level.

b. System flow chart identifying all major modules and illustrating their interaction.

c. All system software in disc image format on magnetic tape.

d. Source code for all system software on magnetic tape.

e. Descriptions of all software modules inclusive of I/O parameters, data passed, function of purpose, external calls, linkages used, pertinent system parameters and formats, status bit assignments, and key flags and controls.

f. Procedures for maintenance and adding instruments, rebuilding operating system and adding computer peripherals.

g. Transform document, describing and defining Compiler Output/Instrument Input, illustrating data flow and transition and identifying all argument passings.

h. Description of all debug aids for software maintenance.

i. Documentation detailing requirements for initiation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

j. Description of all system restrictions, i.e. overlay techniques, memory map adherence, loading, bootstrapping.

k. Benchmark and source data to verify all system software.

l. Operating, maintenance and programming manuals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Hardware and software quality assurance. The equipment and its software shall be subjected to a quality assurance program conforming to the requirements of MIL-Q-9858 and MIL-S-52779 respectively.

4.3 Acceptance tests. The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Representative. Contractors not having adequate facilities for conducting all required tests shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall generate test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance tests are described in the following paragraphs and shall consist of the following:

a. First Article Tests

b. Production Acceptance Tests

4.3.1 First article tests. The contractor shall perform First Article Tests (FAT) in accordance with an approved test plan. The First Article Test shall be performed on the first preproduction unit. The contractor shall provide to the procuring activity for review and approval a detailed FAT Plan that will define and implement the following as a minimum:

a. System Acceptance Test Procedure.

b. Testing to the approved HTS Specification on a paragraph by paragraph basis to verify and identify hardware and software compatibilities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

c. System Self-Test capability verified through fault insertion testing.

d. Identification of all hardware and software required.

e. Conduct a Physical Configuration Audit.

f. Define requirements, format/content, for a final FAT Report.

g. Define a method for closing out identified open items.

4.3.1.1 FAT demonstration. The FAT shall be a contractor to Navy sell-off of the ATE with Navy attendance throughout the entire test proceedings. The FAT shall verify that all interface capabilities are defined and demonstrated in accordance with the approved FAT Plan per 4.3.1. The FAT shall be used to supplement continuous HTS development reviews to ascertain the HTS capabilities and readiness for engineering acceptance.

4.3.1.2 Navy attendance. Representatives of the Navy who are recognized experts in automatic test equipment hardware and software are required to be in attendance at FAT to insure that all FAT results are documented for the purpose of recommending acceptance or rejection to the cognizant contract administration services office.

4.3.1.3 FAT rework/retest. All deficiencies found and identified during the FAT shall be reworked and corrected prior to production acceptance. The rework shall include all required changes to the hardware, software, and all related documentation.

4.3.2 Production acceptance test. Production Acceptance Testing shall be accomplished via an Acceptance Test Procedure and an Acceptance Test Program, generated by the contractor and approved by the Navy, which verifies the operational status of the HTS.

4.3.2.1 Individual test. Each system submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each system accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Burn-In Test

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

4.3.2.1.1 Examination of product. Each system shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.2.1.2 Operational test. Each system shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to ensure satisfactory equipment operation.

4.3.2.1.3 Manufacturing burn-in test. Each system shall be operated under the test conditions specified below for a period of fifty (50) hours without failure. The specified test conditions are as follows:

a. Temperature - Ambient Room

b. Humidity - Ambient Room

c. Vibration - N/A

4.4 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor. The acceptance test procedures shall include verification of the accuracy and completeness of all test programs developed for performance monitoring and end-to end testing, and used to demonstrate specified performance. The right is reserved by the procuring activity or the Government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specificaiton MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

5. PACKAGING

5.1 Product identification and marking. The system, its assemblies, and its subassemblies shall be marked in accordance with source control drawings or manufacturer's drawings. Power and interconnecting cables shall be marked in such a manner as to indicate proper connection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

5.2 Packaging requirements. The packaging of the HTS for delivery shall be in accordance with the requirements of MIL-STD-794. Marking for shipment and storage shall be in accordance with the requirements of MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The Hybrid Test System will be used for functional and diagnostic test of analog and hybrid Units Under Test within a variety of weapons systems at both shore and sea based facilities.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Color Graphics Display Terminal (specified in 3.3.1.5 through 3.3.1.5.3) is an option to be separately offered and would be provided in lieu of the Display Keyboard Terminal specified in 3.3.1.4.

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9 of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9 are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data requirement	Applicable DID
3.2.4	Report, Reliability and Maintainability Program	UDI-R-21131
	Reliability & Maintainability Block Diagrams/Math models	UDI-R-21132
	Report, R/M Allocation	UDI-R-21133
	R/M Prediction	UDI-R-21134
	Report, R/M Status	UDI-R-21137
	Report, failure summary	UDI-R-21139
	Report, R/M Test Plan	UDI-R-21135
	Report, R/M Test Results	UDI-R-21136
	Report, failure	UDI-R-21141A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

Paragraph	Data Requirement	Applicable DID
3.2.7.4	Procedures, test	UDI-T-21347
	Procedures, computer program test	DI-T-2144
	Document, program package (Support Software)	DE-S-2141
3.2.8	Requests, nomenclature and identification requirement and approval (nameplate, drawings)	UDI-E-21336A
3.2.14	System Safety Program Plan (SSPP)	DI-H-7047
	Report, Safety Status	UDI-H-21380
	System Safety Hazard Analysis Report	DI-H-7048
3.3.6	Drawings, engineering and associated lists	DI-E-7031
	Specification, interface design	DI-E-2135
	Data, microelectronic devices and assemblies approval requests	UDI-E-21337
	Specifications, equipment	UDI-E-21342
	Document, data base design	DI-S-2140
	Document, program description	DI-S-2139
3.3.7.4	Manual, Operator's (Support Software)	DI-M-2145
	Procedures, computer program test (Support Software)	DI-T-2114
	Specification, program design (Support Software)	DI-E-2138
	Specification, computer program test	DI-E-2143
4.3	Report, first article (Preproduction) test	UDI-T-21349
	Specification, computer test	DI-E-2143
	Procedures, test	UDI-T-21347

(copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.2.3 Government-furnished property. The Government will furnish the property listed in 3.3.3.5.

6.2.4 First Article Test. When the first article test is required for inspection and approval it shall be tested and approved under appropriate provisions so stated in UDI-T-21349.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

6.2.5 Digital unit. A DWG Interface and Performance Specification, which defines DWG system capabilities and integration requirements, is available to augment this procurement package.

6.2.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:

Navy - AS

(Project 6625-N780)

☆U.S. GOVERNMENT PRINTING OFFICE: 1981-703-023/2279

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-85330A(AS) HYBRID TEST SYSTEM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Standards Department (ESSD)
Code 93
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.